

Error Reasoning Mechanism of Intelligent Correction Experiment Report System Based on Knowledge Graph

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Abstract—This paper introduces the Error Cause Reasoning Mechanism of the knowledge graph-based intelligent grading experiment report system, which takes the database course experiment report as the research object. It aims to solve the problem that the existing Automatic Grading System can only output vague prompts such as “low similarity” and cannot locate specific error causes. By constructing a Domain Knowledge Graph containing Knowledge Point, Typical Expression, Potential Error Type, and Course Module, establishing an Association Paradigm of “low-similarity fragments - Knowledge Point - potential errors”, and designing a Three-level Error Reasoning Algorithm, the transformation from low-similarity text fragments to Knowledge Deficiency is realized. This mechanism can accurately associate experiment report errors with Knowledge Point, provide an effective method for intelligent analysis of learning situations, make up for the shortcomings of existing experimental courses Learning Analytic, and is of great significance for promoting the development of AI-driven Educational Assessment Tool.

Keywords—Learning Analytic, Knowledge Graph, Error Analysis, Intelligent Grading

I. INTRODUCTION

A. Research Background and Motivation

With the deepening of digital transformation of education, the Ministry of Education has clearly proposed to build a three-in-one new teaching ecosystem of “intelligent teaching, intelligent evaluation, and intelligent feedback”. However, most of the current teaching evaluations still follow the traditional manual marking method, which is limited by the teacher's energy and makes it difficult to analyze the knowledge point mastery defects implied in the Experimental Report. On the other hand, the existing Automatic Grading System mostly relies on text similarity comparison, which can only output vague prompts of “low similarity” and cannot locate specific error causes.

B. Research Questions and Objectives

This paper takes the experimental report of database course as the research object, aiming to conduct learning analysis function of the intelligent grading experiment report system driven by Knowledge graph technology. It focuses on how to infer specific error causes and corresponding Knowledge Point from the low-similarity fragments output by the grading

function in the learning analysis function, so as to promote personalized teaching for teachers.

The main goals are as follows:

- Construct a Domain Knowledge Graph for database experiments, and establish the Association Paradigm of “low-similarity fragments-Knowledge Point-potential error”.
- Design a Three-level Error Reasoning Algorithm to realize the transformation from text fragments to Knowledge Deficiency.

II. RELATED WORK

A. Existing Research on Learning Analytic

Traditional Learning Analytic mostly relies on teachers' subjective experience, inferring students' learning status through methods such as classroom observation and homework marking, which suffers from problems such as being experience-based and one-sided[1][2]. With the digital transformation of education, technology-driven approaches have become the mainstream research direction of Learning Analytic: supported by big data and strong algorithms, it enables comprehensive collection, efficient analysis, and real-time feedback of learning data. For example, sentiment analysis of students' text materials is performed through natural language processing[3], computer vision is used to recognize classroom engagement[1], or student cognitive models are constructed using learning behavior data[4], which significantly improves the objectivity and accuracy of the analysis.

Existing Learning Analytic has been widely applied in various disciplines and teaching links in basic and higher education, ranging from ideological and political courses[5][6], information literacy courses[7] to junior high school geography[8]. However, the current Learning Analytic system often attributes error points to superficial reasons, lacking correlation analysis with specific Knowledge Point, making it difficult to accurately locate learning obstacles[9][2]; and the scenario adaptability is limited, especially in experimental courses, there is less Learning Analytic for operation process, report standardization, etc., which is difficult to meet the needs of practical teaching. This research conducts Learning Analytic

based on the Automatic Grading System of database experiments reports. Based on existing research, it specifically addresses the practical scenario of database experiments reports, making up for the shortcomings of existing research in experimental courses Learning Analytic, and improves the scientific of attribution analysis through accurate matching of error points and Knowledge Point.

B. Application of Knowledge Graph Technology in Education

Knowledge graph is an efficient way of knowledge organization[10], which can help users quickly obtain the information or knowledge they need. The application of knowledge graph in the field of education is mainly to construct curriculum knowledge graphs. According to the characteristics of the curriculum, graph technology is used to construct knowledge graphs that meet various teaching tasks[11], integrating teaching content, teaching resources, and teaching objectives, focusing on curriculum system resource integration and visualization[12].

At present, Knowledge graph technology is mainly used for the integration of knowledge systems, which cannot reflect students' mastery of knowledge and cannot be directly applied to learning situation analysis.

This study constructs a knowledge graph with a "Knowledge Point - Typical Expression - potential error" three-layer structure, leveraging the characteristics of Knowledge graph technology for rapid information retrieval. It transforms low-similarity ambiguous results present in Experimental Report into interpretable error points, closely associating problems with Knowledge Point. This enables Learning Analytic based on the mastery of each Knowledge Point, better facilitating the implementation of teaching activities.

C. Limitations of Existing Solutions

Currently, Knowledge graph technology is not widely used in Learning Analytic, but more in building curriculum knowledge graphs and integrating knowledge systems. At present, Learning Analytic focuses more on the overall performance of primary and secondary school students in basic subjects or ideological and political courses, including learning attitudes and learning conditions. There is less research on Learning Analytic for university practical courses, especially using Experimental Report for Learning Analytic.

III. ERROR REASONING MECHANISM FOR EXPERIMENT REPORTS BASED ON KNOWLEDGE GRAPH

This chapter will detail the construction design of the Knowledge Graph that can identify Experimental Report error points and its error reasoning mechanism.

A. Construction Design of Knowledge Graph

The Knowledge Graph aims to establish a reasoning intermediary of "low-similarity fragments-Knowledge Point-potential error" to solve the problem that the system can only output "low similarity" but cannot directly identify error points. It has 3 key construction processes.

- Knowledge Acquisition and Refinement: Extract the core concepts of the database foundation course from teaching resources such as course textbooks and teacher lesson plans, identify entities and their relationships through natural language processing technology, and construct a structured knowledge framework combined with the experience of domain experts.
- Do Entity and Relationship Design: We introduce "Typical Expression" as a bridge connecting similarity fragments and Knowledge Point, and define three entities: Knowledge Point, Typical Expression, and Potential Error Type. Simultaneously, the Course Module entity is defined as the upper-level classification of Knowledge Point. The specifics are shown in Table 1. The relationships between entities are defined by triples, with the core relationships including:
 ("Knowledge Point", "contains Typical Expression", "Typical Expression")
 ("Knowledge Point", "associated potential error", "Potential Error Type")
 ("Course Module", "contains Knowledge Point", "Knowledge Point")
- Storage Structure Design: We use the Neo4j graph database to store entities and relationships [13], and design a two-layer reasoning mechanism of "Course Module-Knowledge Point" and "knowledge - standard - error" to realize the mapping from "low-similarity fragments" to "potential errors".

The application of Knowledge Graph in education aims to integrate complex knowledge and provide support for error-prone reasoning. Existing research [14] indicates that it can associate knowledge from different fields to help understand the causes and solutions of knowledge. In this database experiments scenario, we further expand its application scope and reasoning depth to adapt to the grading requirements based solely on Similarity comparison.

TABLE I. ENTITY TYPES AND THEIR ROLES

Entity type	Definition and function	example
knowledge point	Course core concepts and skills	INSERT statements, foreign key constraints
Typical expression	standard representation of knowledge points	"FOREIGN KEY (column) REFERENCES primary_table (column)"

Potential error types	Common mistakes related to knowledge points	Number of fields mismatch, REFERENCES clause not specified.
Course module	The superordinate classification of Knowledge Point, used to organize the entire course's knowledge systems.	Data Definition Language (DDL) module, Data Manipulation Language (DML) module, Transaction Management module.

B. Implementation Algorithm for Reasoning from Low Similarity to error causes

To address the issue of the system only outputting "Report segment has low Similarity with the answer," a 3-step error causes reasoning process is designed. The overall reasoning process is shown in Fig. 1.

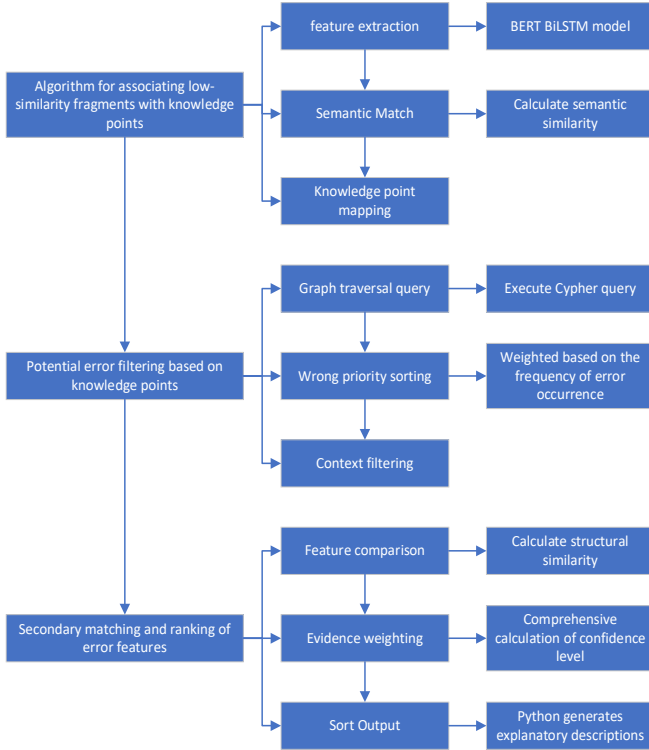


Fig. 1. Error Cause Reasoning Flowchart

1) *Algorithm for associating low-similarity fragments with knowledge points*: When the system detects a low similarity between a fragment and the standard answer, it extracts keywords from the fragment and performs semantic matching with the "Expression Content" of the "Typical Expression" node. If the matching is successful, it is associated with the corresponding "Knowledge Point." Let S represent the detected low-similarity report fragment, G represent the knowledge graph, and K represent the set of associated knowledge points. The specific algorithm is as follows:

a) *Feature extraction*: The BERT-BiLSTM model is used to accurately extract the key features of fragment S through bidirectional semantic encoding. Then, core operation words (e.g., SELECT, JOIN, WHERE) and object entities (table names, field names) are extracted.

$$f_s = \text{BiLSTM}(\text{BERT}(S)) \quad (1)$$

b) *Semantic matching*: Calculate the semantic similarity with the "Typical Expression" nodes in the knowledge graph. If $\text{sim}(s, t_i) > \theta_t$ (the threshold is set to 0.85), the association is activated. $t_i \in T$ (T is the set of typical representations)

$$\text{sim}(s, t_i) = \cos(f_s, f_{t_i}) \quad (2)$$

c) *Knowledge point mapping*: Knowledge points are determined through the graph relationship $R(t_i \text{ belongs to } k_j)$.

$$K = \{k_j \mid \exists t_i \in T, R(t_i, k_j) \wedge \text{sim}(s, t_i) > \theta_t\} \quad (3)$$

2) *Potential error filtering based on knowledge points*: Through the relationship of "knowledge point - associated potential error" in the knowledge graph, filter all potential error types under the knowledge point. Let K represent the knowledge point set, G represent the knowledge graph, and E represent the potential error type set. The specific algorithm is as follows:

a) *Graph traversal query*: For each $k_j \in K$, execute the Cypher query.

b) *Error priority ranking*: Based on error occurrence frequency weighting, where $\alpha = 0.7$ and $\beta = 0.3$. The formula is as follows:

$$\text{weight}(e_i) = \alpha \cdot \text{freq}(e_i) + \beta \cdot \text{severity}(e_i) \quad (4)$$

c) *Context filtering*: Retain errors that match the current operation type, the formula is as follows, where $\text{ops}(S)$ is the operation type in the segment.

$$E = \{e_i \mid \text{type}(e_i) \in \text{ops}(S)\} \quad (5)$$

3) *Secondary matching and ranking of error features*: For the filtered potential error type, extract the features of the student's segment and perform a secondary match with the "error features" of the "potential error type" node, and output the results sorted by the matching score. Let S represent the student's segment and E represent the potential error set. Finally, output the sorted error diagnosis results. The specific algorithm is as follows:

a) *Feature comparison*: Extract the error feature triplet $\varphi(e_i) = \{\text{pattern}, \text{context}, \text{constraint}\}$, and then calculate the structural Similarity, the calculation formula is as follows:

$$\text{match_score} = \frac{|\text{AST}(S) \cap \text{AST}(\Phi(e_i))|}{|\text{AST}(\Phi(e_i))|} \quad (6)$$

b) Evidence weighting: Combine syntactic and semantic evidence weighting for confidence synthesis calculation.

$$\text{conf}(e_i) = 0.6 \cdot \text{syntax_match} + 0.4 \cdot \text{semantic_match} \quad (7)$$

c) Sorted output: First, sort in descending order of confidence (as shown in the formula below), and then use Python to generate explanatory descriptions.

$$\text{Diagnosis} = \{ (e_i, \text{conf}(e_i)) \mid \text{conf}(e_i) > 0.75 \} \quad (8)$$

IV. DISCUSSION

A. Research Contributions

This research makes the following innovative contributions in the field by driving the intelligent grading experiment report system's learning analysis function with Knowledge graph technology:

- Constructing a domain knowledge graph for database experiments, establishing an association paradigm of “low-similarity fragments - knowledge point - potential error”, and achieving precise association between experiment report errors and knowledge point.
- Designing a three-level error reasoning algorithm that progresses from associating low-similarity fragments with knowledge point, to filtering potential error based on knowledge point, and then to secondary matching and ranking of error features, completing the transformation from text fragments to knowledge deficiencies, and providing an effective method for intelligent analysis of learning situations.
- Addressing the practical scenario of database experiments reports, this work fills the gap in existing research on learning analysis in experiment-based courses, enhances the scientific rigor of error attribution analysis, and helps promote personalized teaching for teachers.

B. Limitations and Future Work

Despite the progress made in error cause reasoning for the knowledge graph-based intelligent grading experiment report system, there are still some limitations:

- **Limitations of the reasoning chain:** It relies entirely on the linear, unidirectional reasoning chain of “low-similarity fragments → knowledge point → potential error,” which has logical breaks. When the initial association fails, subsequent error reasoning will be completely halted, resulting in a lack of flexibility in the framework.
- **Limitations of application scenarios:** Currently, it is mainly aimed at experiment reports for the database course, with a narrow application scope. It has not yet been verified in the grading of experiment reports for other disciplines.

- **Completeness of the knowledge graph:** The construction of the knowledge graph relies on resources such as course textbooks and teacher lesson plans, and there may be issues such as incomplete coverage of knowledge point and untimely updates of potential error type.

Future research can be carried out in the following aspects:

- Optimize the reasoning mechanism: Introduce multi-path reasoning or fault-tolerant mechanisms to solve the logical fracture problem of the current linear reasoning chain and improve the flexibility and robustness of the framework.
- Expand application scenarios: Apply this Error Reasoning Mechanism to the grading of lab reports in other disciplines to verify its universality and make adaptive adjustments according to the characteristics of different disciplines.
- Improve the knowledge graph: Establish a dynamic update mechanism to timely supplement new knowledge point and potential error type to ensure the timeliness and comprehensiveness of the knowledge graph and improve the accuracy of error reasoning.

V. CONCLUSION

This study takes the experiment reports of database course as the research object. By constructing a domain knowledge graph for database experiments and designing a three-level error reasoning algorithm, it realizes the reasoning from low-similarity fragments output by the grading function to the specific error causes and corresponding knowledge point in the intelligent grading experiment report system, providing an effective method for intelligent analysis of learning situations.

The research results can help teachers accurately locate students' deficiencies in knowledge mastery, promote the development of personalized teaching, and also make up for the shortcomings of existing learning analysis in experimental courses. In the future, through the improvement and optimization of limitations, it is expected to further improve the applicability and effectiveness of the Error Reasoning Mechanism, and provide strong support for intelligent evaluation and feedback in the context of digital transformation of education.

REFERENCES

- [1] Zeng YH, Peng C. AI-supported learning analysis: Dilemma and Path [J]. Journal of Hebei Normal University (Educational Science Edition), 2025, 27(03):82-88.
- [2] Liu G, Tian J. Value Implication, Practical Dilemma and Improvement Path of learning analysis [J]. Teaching and Management, 2020, (27):18-21.
- [3] Pu QP, Wang XT. Theoretical Framework and Practical Path of Artificial Intelligence Assisted learning analysis [J]. University Education Science, 2024, (03):31-38.
- [4] Zhu BJ, Shi TN, Shi ZJ, Zhu JJ, Xu JL, Zhu L, Yang W, Xie WM, Wu WH. Construction of "Pyramid" Network Learning Platform and Learning Behavior Analysis [J]. Experimental Technology and Management, 2021, 38(08):208-212+216.

- [5] Chen AX, Yao LM. The Logical Path and Promotion Strategy of Ideological and Political Theory Course learning analysis[J]. Ideological & Theoretical Education, 2023,(04):75-79.
- [6] Deng Y. System Innovation of Ideological and Political Courses in Colleges and Universities in the New Era from the Perspective of "Student-centered" learning analysis[J]. Journal of National Academy of Educational Administration, 2025,(03):77-86.
- [7] M. Young, The Technical Writer's Handbook. Mill Valley, CA: University Science, 1989.
- [8] Yu C D. Design of Junior High School Geography teaching activities Based on learning analysis: Taking "Natural Disasters" as an Example [J]. Teaching Reference for Middle School Geography, 2023,(03):59-61.
- [9] Wang Q, Jin Y. A Review of learning analysis Research in Secondary Education Stage [J]. Educational Theory and Practice, 2023,43(02):54-57.
- [10] Li T, Wang CC, Li H K. Development and Construction of Knowledge Graph [J]. Journal of Nanjing University of Science and Technology, 2017,41(01):22-34.DOI:10.14177/j.cnki.32-1397n.2017.41.01.004.
- [11] Feng Y, Liu X Y, Li K X, Sun X. Survey on Automatic Construction of curriculum knowledge graphs [J]. Computer Technology and Development, 2025,35(1):1-11.
- [12] Wang H X, Yang Y, Xia S L, Wang L, Li P H. Research on Knowledge Graph Empowering the Characteristic Curriculum of Transportation Engineering—Innovative Path and Practical Exploration [J]. Highway Traffic Technology, 2025,25(3):106-112
- [13] Miller J J. Graph Database Applications[M]. O'Reilly, 2022: 78-82.
- [14] Wang SS, Li M. Reasoning and Diagnosis of Error-Prone Points Based on Knowledge Graph[J]. Computer Engineering and Applications, 2022, 58(15): 231-238.